CS6037 Machine Learning Assignment 4

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Part 1 – Programming

P1 – The VC Confidence Term

P2 – A Shattering Checker

a)

b)

Shatterability in Dimension

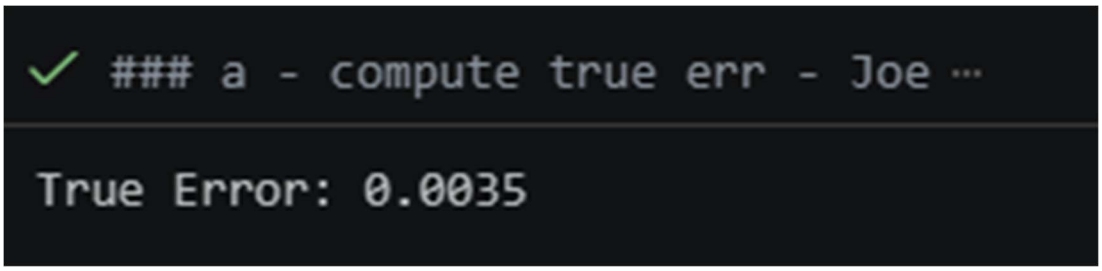
N	N+1 Points	N+2 Points
2.0	1.0	0.0
3.0	1.0	0.0
4.0	1.0	0.0
5.0	1.0	0.0

c)

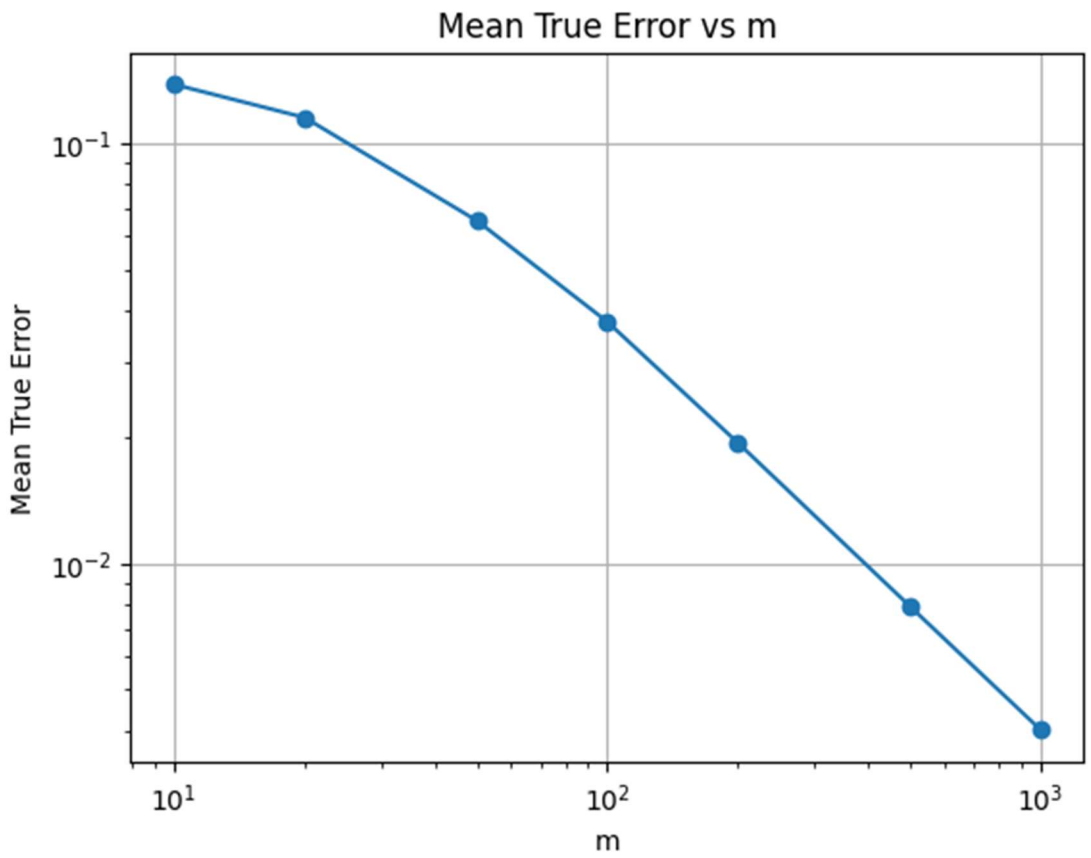
d)

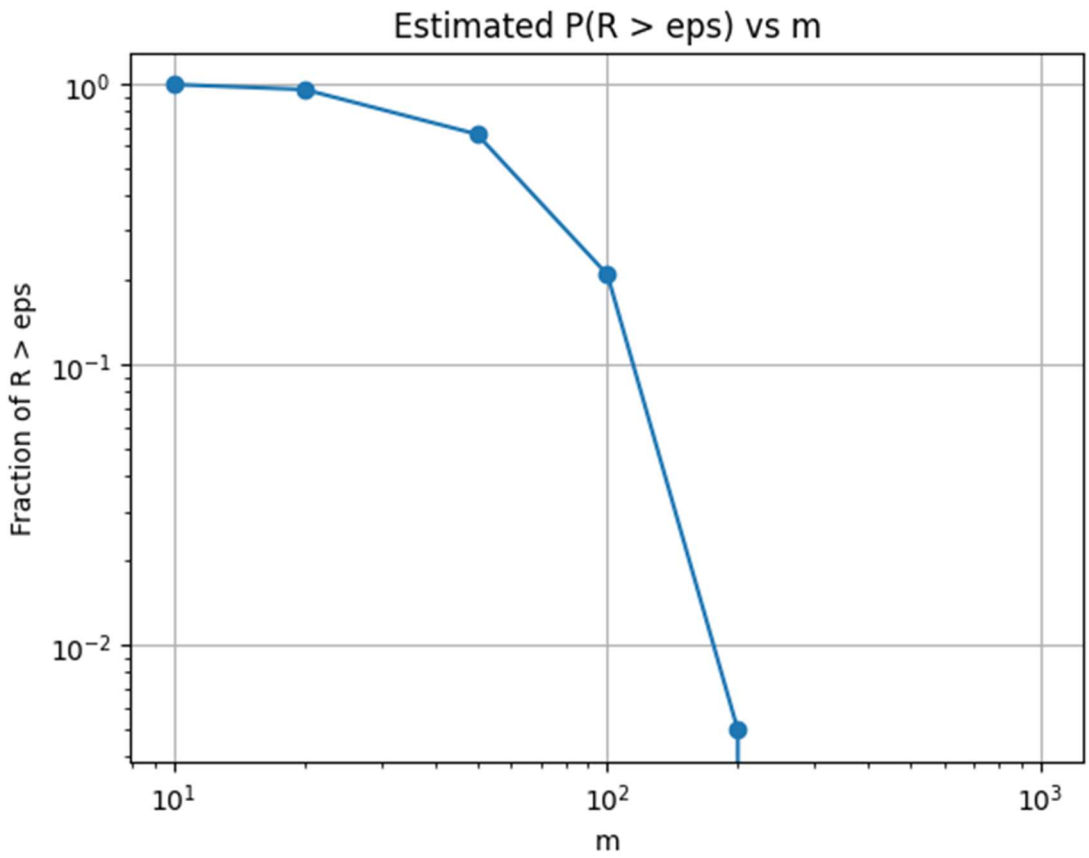
P3 - PAC learning in action: rectangles and thresholds

a)

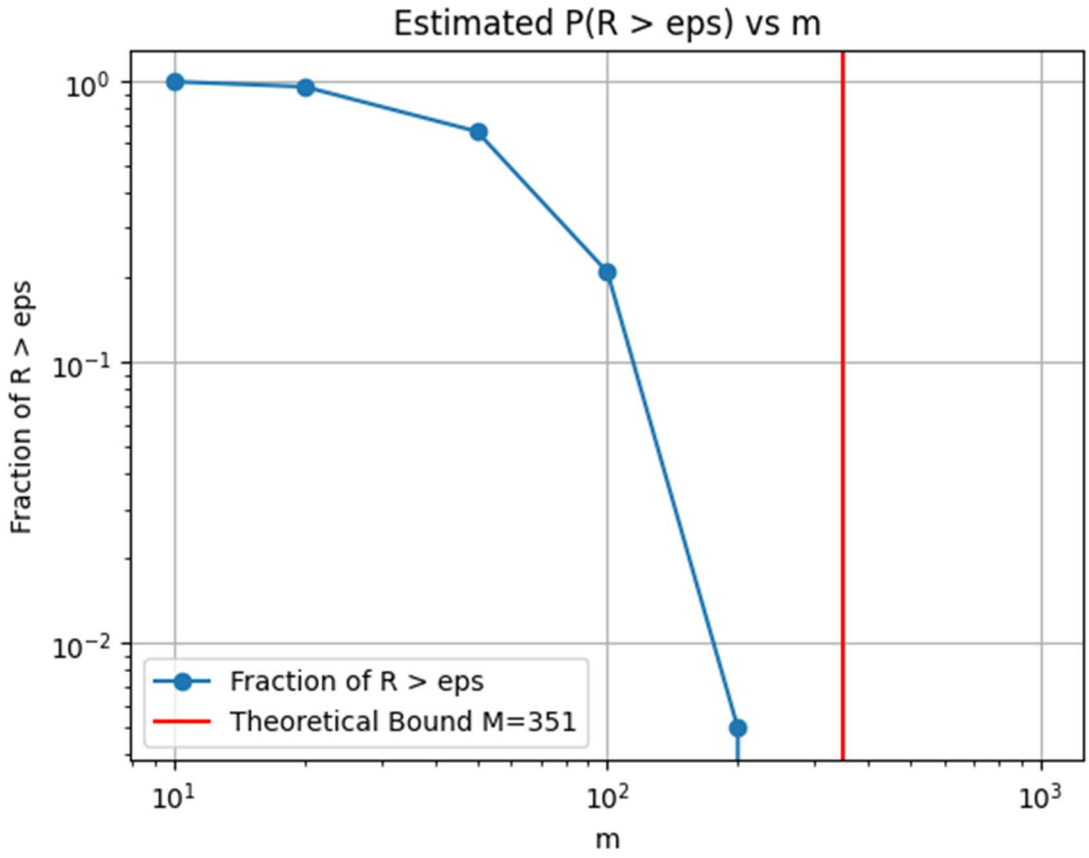


b)


Mean True Error vs m
Mean True Error
10^{-1}
10^{-2}
10^{1}
10^{2}
10^{3}
m


Estimated P(R > eps) vs m
Fraction of R > eps
m
10^{0}
10^{-1}
10^{-2}
10^{1}
10^{2}
10^{3}

c)



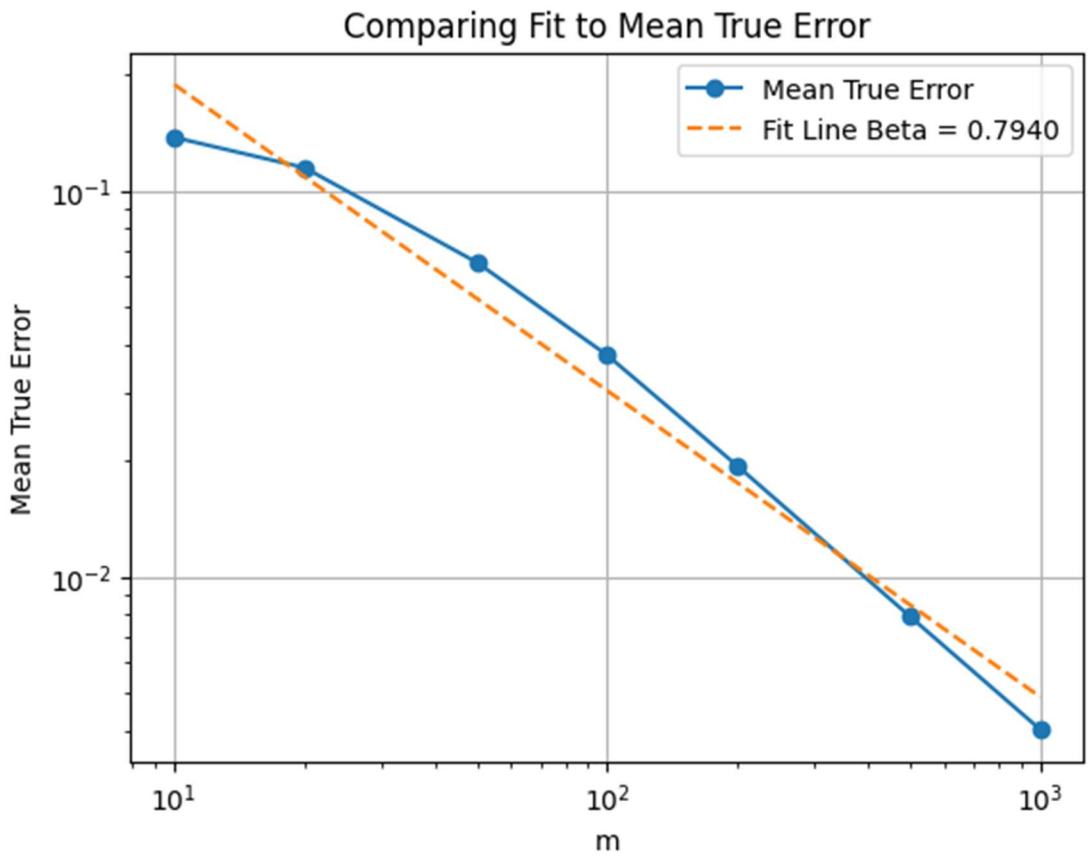

```
✓ ### c - find smallest m (to nearest 10) - Joe ···

Lowest M - to 10s : 150
Empirical Probability: 0.05000
Pessimistic Factor: 2.34000
```

d)

```
✓ ### d - fit straight line of mean vs err = Joe ···

Beta : 0.7940
```

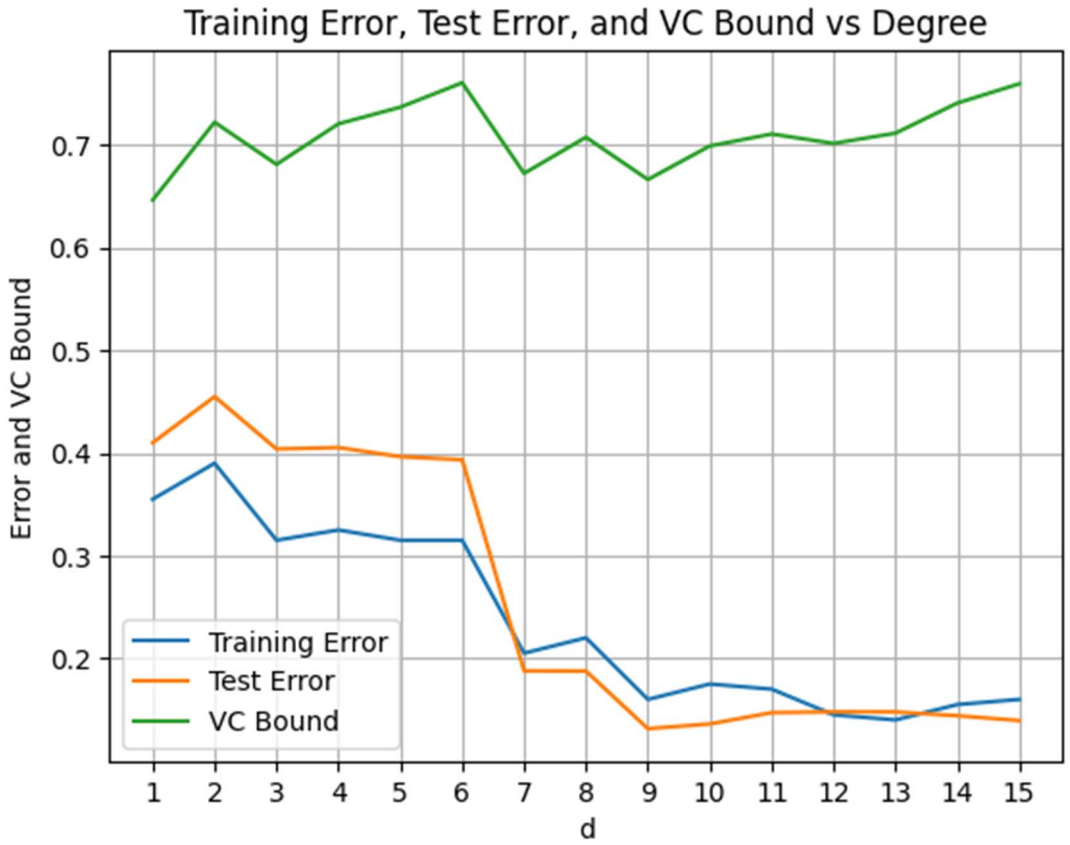


P4. Structural Risk Minimization versus cross-validation

a)

b)

c)



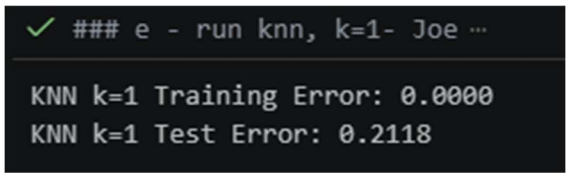

```
✓ ### c - finding which degree SRM selects - Joe ···
Degree selected by SRM: 1
```

```
✓ ### c - finding which degree 5-fold cross validation selects - Joe ···
Degree selected by 5-fold CV: 15
```

```
✓ ### c - find which minimizes test error- Joe ···
Degree selected by minimum test error: 9
```

d)

e)



Part 2 – Essay – Prompt C